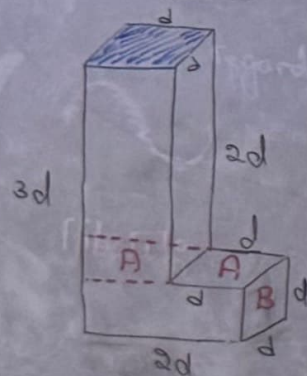


ex. 2

A gauge pressure needed to suck a fluid of density ρ up to h :

$$P_{\text{suck}} = -\rho g h = -1800 \times 9.8 \times 1.5 = -26460 \text{ Pa}$$

ex. 3



L-shaped tank contains water open at the top if $d = 5 \text{ m}$.

① The force on face A is:

$$h_A = 2d = 2 \times 5 = 10 \text{ m}, A_A = d \times d = 5 \times 5 = 25 \text{ m}^2$$

$$P_{\text{gauge}} = \rho g h = 1000 \times 9.8 \times 10 = 98000 \text{ Pa}$$

$$P_t = \rho g h + P_{\text{atm}} = 98000 + 10^5 = 198000 \text{ Pa}$$

$$F_t = P_t \times A = 198000 \times 25 = 4950000 \text{ N}$$

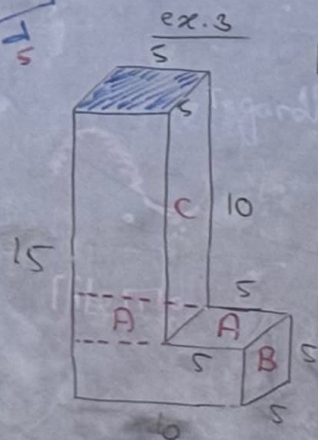
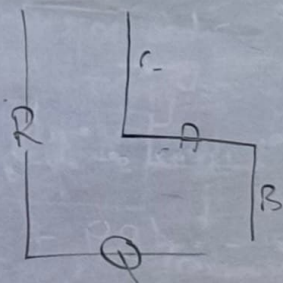
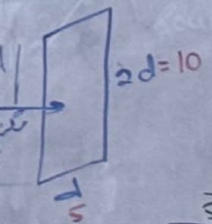
② The force on face B is:

$$h_B = 10 + 2.5 = 12.5 \text{ m}, A_B = d \times d = 5 \times 5 = 25 \text{ m}^2$$

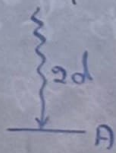
$$P_{\text{gauge}} = \rho g h = 1000 \times 9.8 \times 12.5 = 122500 \text{ Pa}$$

$$P_t = \rho g h + P_{\text{atm}} = 122500 + 10^5 = 222500 \text{ Pa}$$

$$F_t = P_t \times A = 222500 \times 25 = 5562500 \text{ N}$$



L-shaped tank contains water open at the top, if $d = 5\text{ m}$.



① The force on face A is:
 $h_A = 2d = 2 \times 5 = 10\text{ m}$, $A_A = d \times d = 5 \times 5 = 25\text{ m}^2$

$$P_{\text{Gauge}} = \rho g h = 1000 \times 9.8 \times 10 = 98000\text{ Pa}$$

$$P_t = \rho g h + P_{\text{atm}} = 98000 + 10^5 = 198000\text{ Pa}$$

$$F_t = P_t \times A = 198000 \times 25 = 4950000\text{ N}$$

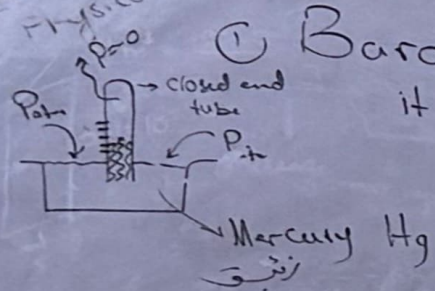
② The force on face B is:

$$h_B = 10 + 2.5 = 12.5\text{ m}$$

$$P_{\text{Gauge}} = \rho g h = 1000 \times 9.8 \times 12.5 = 122500\text{ Pa}$$

$$P_t = \rho g h + P_{\text{atm}} = 122500 + 10^5 = 222500\text{ Pa}$$

$$F_t = P_t \times A = 222500 \times 25 = 5562500\text{ N}$$



① Barometer

it is used to measure the atmospheric pressure.

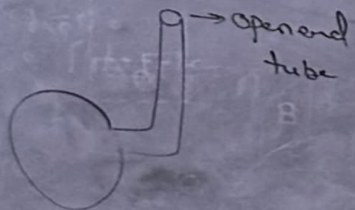
fluid in a tube is Hg $\rightarrow h = 760 \text{ mm} = 76 \text{ cm} = 0.76 \text{ m}$

$$P_{\text{atm}} = 760 \text{ mm Hg}$$

$$P_{\text{atm}} = \rho g h$$

10^5 Pa

② Manometer



it is used to measure the fluid pressure.

$$P_{\text{gauge}} = P_{\text{abs}} - P_{\text{atm}} = \rho g h$$